

Distributed Detection with Multiple Sensors: Part II—Advanced Topics

RICK S. BLUM, MEMBER, IEEE, SALEEM A. KASSAM, FELLOW, IEEE,
AND H. VINCENT POOR, FELLOW, IEEE

Invited Paper

Following the foundational work that established basic ideas for optimum distributed detection schemes using multiple sensors (as reviewed in Part I of this two-part review), further work on distributed detection has developed many useful and interesting extensions of the basic concepts. These more recent developments parallel those that arose from the early work on centralized, classical signal detection, resulting in new ideas of asymptotically optimum, nonparametric, robust, and sequential centralized detection. Recent developments on these topics in the setting of distributed signal detection are reviewed in the present paper. Results in these directions are important in practice because they allow cases of modeling uncertainty to be addressed, and they provide more efficient detection schemes by optimizing more general performance criteria.

I. INTRODUCTION

In Part I of this paper [1] the basic ideas of distributed signal detection were discussed. In particular, Part I focussed on optimality under the Neyman–Pearson and Bayesian detection criteria, which lead in standard situations to likelihood ratio tests at the individual sensors and at the fusion center of a distributed detection network. The optimum schemes for distributed signal detection under these criteria satisfy a set of coupled constraints, and while the optimization under such coupled conditions is not trivial and leads to considerable subtleties, nonetheless we may view such results as the counterparts of classical centralized signal detection schemes. By classical signal detection schemes, we mean techniques derived for binary hypothesis testing models under parametric, usually Gaussian,

statistical models for noise and signals, obtained for fixed-sample-size detectors under the Neyman–Pearson, Bayes, or minimax criteria, with particular interest in optimum processing for deterministic coherent and narrowband signals and for random signals. The systematic study of classical centralized signal detection began approximately 50 years ago, and the corresponding fundamentals of its distributed-sensors version have become a focus of work in signal detection in the last two decades.

These basic concepts also form the foundation for further work on distributed signal detection under other related performance criteria, with fewer assumptions on the inputs, and for modified test structures within the distributed sensor framework. In this paper, the second part of the two-part review and survey of results on distributed signal detection contained in this issue, we will examine some of these further developments. In particular we will review work, which in most cases is fairly recent, on optimum distributed detection for weak signals, on nonparametric and robust distributed detection, and on sequential schemes for distributed detection. One finds in these further results on distributed detection a clear correspondence with the evolution of interest in centralized signal detection from classical parametric problems to studies of locally optimum detection [2], [3], nonparametric detection [4]–[7], and sequential [3], [8] detection.

When a simple hypothesis testing problem in signal detection is expanded to a composite hypothesis problem, one would like to be able to obtain uniformly most powerful detection results for an optimum scheme. A simple but important example of this occurs when we can no longer assume that the amplitude of a deterministic signal in additive noise is precisely known when forming the detection test statistic. Unfortunately it is generally not possible to find uniformly most powerful solutions in such cases, especially for distributed detection and under non-Gaussian noise conditions. The theory of locally optimum detection in centralized situations offers an approach that can be taken

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R. S. Blum is with the Electrical Engineering and Computer Science Department, Lehigh University, Bethlehem, PA 18015 USA (e-mail: rblum@eecsl.lehigh.edu).

S. A. Kassam is with the Department of Electrical Engineering, University of Pennsylvania, Philadelphia, PA 19104 USA (e-mail: kassam@pender.ee.upenn.edu).

H. Vincent Poor is with the Electrical Engineering Department, Princeton University, Princeton, NJ 08544 USA (e-mail: poor@princeton.edu).

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to design good schemes. In particular, this theory focuses on the asymptotic case of signals of vanishing strength so that performance is improved for the most challenging weak-signal detection situation. In Section II of this paper we review some recent results on locally optimum distributed detection based on a generalized Neyman–Pearson criterion, with specific discussion of random signals in a two-sensor distributed scheme and of narrowband signal detection. The noise models used here are not necessarily Gaussian, and the signal components at the different sensors are not necessarily independent, and these conditions lead to some interesting and apparently nonintuitive structures for the optimum sensor and fusion processors.

In Section III we examine work that has been reported on nonparametric distributed detection as well as on what are known as constant false-alarm rate (CFAR) distributed detection schemes. In nonparametric detection, the classical requirement of parametric statistical models for the signal and noise observations is relaxed and the competing composite hypotheses are stated in terms of statistical models for the observations that are not tied down by probability density function models from specific parametric families. Schemes based primarily on signs and ranks of observations, which have provided very useful structures for nonparametric detection in the centralized structure, have found interesting applications in distributed detection also. CFAR schemes are generally based on parametric hypotheses with often a noise power parameter unknown. The parameters in such a model are estimated from available observations to obtain fixed or bounded false-alarm probability, with particular attention to performance under abrupt changes in, or nonhomogeneity of, noise power. Robust signal detection theory based on the concepts of minimax detection performance over composite classes of probability density functions has produced very useful structures for centralized detection. Such techniques are of value when precise assumptions about a nominal statistical model must be tempered to allow some degree of deviation to more accurately reflect conditions occurring in practice. These ideas can be and have been extended to the distributed sensors setting, and are briefly reviewed in Section IV.

Sequential hypothesis testing has been of great interest in statistics and also in signal detection for many years. In centralized schemes for signal detection, sequential testing offers the possibility of making final decisions within a given reliability requirement as soon as enough data has been collected to stop further data acquisition and declare a result. Sequential procedures are useful when data acquisition is costly and when both reliability and decision delay are important considerations. In much more recent work these ideas have been applied in distributed detection, where sequential procedures may be implemented at the individual sensors or at the fusion center. Section V contains a review of this recent work, and also discusses recent results on quickest distributed change detection. Change detection is of interest in monitoring applications, where it is important to identify, as soon after it occurs as is

possible, any change in the distribution from which data is being drawn.

Detection of weak signals and detection under composite hypotheses in general, under conditions of non-Gaussian noise, are clearly of interest in applications. Nonparametric and robust techniques provide detection structures that are often simple and perform well under conditions likely to occur in reality. Sequential detection allows detection decisions to be made reliably without unnecessary delays when decision delay and data acquisition cost are important factors. All of these ideas, when combined with distributed detection using data from isolated sensors, form a powerful set of techniques that are of great interest in applications ranging from machinery monitoring to large-scale radar and sonar detection and tracking systems.

II. LOCALLY OPTIMUM DISTRIBUTED SIGNAL DETECTION

Consider the design of an N -sensor distributed detection scheme, which is to decide between a signal-present alternative hypothesis and a simple null hypothesis. Each sensor has an associated processor that makes a decision based only on the observations obtained from its local sensor. The sensor processors transmit their decisions to a single central fusion center where an overall decision is made. A particular value y_k of the random vector $\underline{Y}_k$ is observed at the k th sensor, $k = 1, \dots, N$, where y_k is a vector of m_k real observations. In general, the $\underline{Y}_1, \dots, \underline{Y}_N$ may not be independent. Let the overall binary decision be denoted by the random variable U_0 , with a particular realization of U_0 denoted by u_0 and where $u_0 = 0$ corresponds to a signal-absent decision and $u_0 = 1$ corresponds to a signal-present decision. Similarly, U_k is the binary random variable with realization u_k that describes the decision made at the k th sensor.

It is often the case that the alternative hypothesis is not simple; in particular the signal strength may not be known. Assume then that the probability density function (pdf) of the observations at the k th sensor $f_{Y_k}(y_k | \theta)$ is parameterized by a signal-to-noise ratio parameter θ and that the alternative hypothesis corresponds to the case where $\theta > 0$ ($\theta \neq 0$ can also be considered) while the null hypothesis corresponds to $\theta = 0$ (signal absent). In such cases the detection probability $P_d(\theta) = \text{Prob}(U_0 = 1 | \theta)$ will generally depend on θ . In many cases one can not find a uniformly most powerful test that will maximize the probability of detection for all possible θ within a fixed bound on the false-alarm probability, $P_f = \text{Prob}(U_0 = 1 | \theta = 0)$. In these cases one can search for a locally optimum (LO) scheme [2], [3], [4], [9]–[11] that maximizes $\frac{d}{d\theta} P_d(\theta)|_{\theta=0}$ under the constraint of fixed false-alarm probability $P_f = \alpha_0$, assuming $\frac{d}{d\theta} P_d(\theta)|_{\theta=0}$ is not identically zero for all admissible sensor tests. Clearly, LO schemes optimize, in a specific sense, performance for weak signals. If $\frac{d}{d\theta} P_d(\theta)|_{\theta=0}$ is identically zero (this will be the case for some problems involving zero-mean random signals in additive noise, for example) then an LO scheme maximizes $\frac{d^\ell}{d\theta^\ell} P_d(\theta)|_{\theta=0}$ for the lowest order ℓ for which

this is not identically zero for all admissible tests [2]. Let ℓ be defined in this manner for the rest of this paper.

An LO distributed detection scheme was considered in [12] as a suboptimum space-diversity combining technique for a bandpass communications problem. In [12], coherent detection of an on-off keying signal with amplitude fluctuations was considered for the case where the signal is observed in additive non-Gaussian noise. Later, LO distributed detection of known and random signals was studied in [13] for cases with additive non-Gaussian noise. In [13], the optimum sensor decision rules are found for schemes which use fixed fusion rules of a specific type. The fusion rules considered in [12] and [13] include the AND and OR rules and the results are limited to the case where the observations are *independent* from sensor to sensor. These results indicate that, for independent observations, the best sensor rules are of the same form as those used by the optimum isolated sensor scheme with the same statistical model for its observations. The best isolated sensor tests have been studied under the topic of centralized detection theory [2], [3], [9]–[11].

The design of an optimum fusion rule for a fixed set of sensor decision rules is in principle straightforward. Since this is a centralized problem, the LO fusion rule is obtained as the comparison of the ℓ th derivative of the likelihood ratio of the sensor decisions to a threshold. For cases with statistically dependent observations from sensor to sensor, the optimum sensor tests are more interesting, since they are generally different from centralized tests. Given a fusion rule and assuming the rules of all other sensors are also given, it has been shown [14] that the LO sensor test statistic at the k th sensor must be of the form (see (1) at the bottom of the page). Here $\underline{u}^k$ is a set of sensor decisions at all sensors except at sensor k , the sums in the second term of (1) are over all values of $\underline{u}^k$ and $\text{Prob}(\underline{U}^k = \underline{u}^k | \underline{Y}_k = \underline{y}_k, \theta)$ is the conditional probability of making a particular set of sensor decisions at all but the k th sensor given a specific set of observations at sensor k . The quantity $C(\underline{u}^k) = \text{Prob}(U_0 = 1 | \underline{U}^k = \underline{u}^k, U_k = 1) - \text{Prob}(U_0 = 1 | \underline{U}^k = \underline{u}^k, U_k = 0)$ where $\text{Prob}(U_0 = 1 | \underline{U}^k = \underline{u}^k, U_k = u_k)$ depends only on the fusion rule. The test statistic T_k in (1) is the sum of two terms. The first term is the derivative of the likelihood ratio of the observations at the k th sensor, which is the optimum test statistic for an isolated sensor. If the second term in (1) depends on $\underline{y}_k$, this can make the optimum test statistic at the k th sensor different from that which is optimum for an isolated sensor. In the case of independent observations, the quantities $\text{Prob}(\underline{U}^k = \underline{u}^k | \underline{Y}_k = \underline{y}_k, \theta)$ will not depend on $\underline{y}_k$, and so for this case the sensor test statistics that are best for isolated sensors will be best for the distributed scheme.

The optimum sensor test statistic in (1) is compared to a threshold λ_k to make a final decision.

When considering weak-signal detection problems, large sample sizes are frequently of interest. In such cases asymptotic performance measures become appropriate. Asymptotic performance measures focus on the limiting case where the number of observations approaches infinity. For cases with independent observations from sensor to sensor, distributed detection of weak signals based on an asymptotically large number of observations has been studied, for example, in [15]–[18]. In [15], the performance of two structures is compared. The first uses a finite number of sensors, each collecting an asymptotically large set of observations. The second employs asymptotically many sensors each collecting a single observation. Cases with an asymptotically large number of sensors are also considered in [16] and [17]. Distributed schemes without feedback are discussed in [16], while [17] considers schemes with feedback. An approximate performance measure is considered in [18] for cases with an asymptotically large number of observations at each sensor. For cases with correlated observations, distributed detection of weak signals based on an asymptotically large number of observations at each sensor has been studied in [19]–[22], for example. The focus in [19] and [20] is on cases with binary sensor decisions, while [21] and [22] consider cases where multibit sensor decisions are allowed. There has also been extensive investigation of asymptotic performance for nonweak signal cases. For a sampling of this literature see [23].

A. Locally Optimum Random Signal Detection with Two Sensors

We now examine a case with a more concrete observation model, in which a random signal is observed in additive, possibly non-Gaussian, zero-mean noise. Assume that we have the same number of observations at each sensor, $m_1 = m_2 = m$ and that the m observations at the first sensor can be modeled as

$$Y_{1i} = \theta S_{1i} + W_{1i}, \quad i = 1 \cdots m. \quad (2)$$

Here $\underline{S}_1 = (S_{11}, \dots, S_{1m})$ is the vector of random signal samples and the W_{1i} , $i = 1, \dots, m$, are the independent, identically distributed (i.i.d.) noise samples that have a common univariate pdf f_{W_1} . The m observations at the second sensor are assumed to obey

$$Y_{2i} = \theta S_{2i} + W_{2i}, \quad i = 1 \cdots m \quad (3)$$

with $\underline{S}_2$, S_{2i} , and W_{2i} similarly defined. The noise observations are assumed to be independent from sensor to sensor, but the signal observations may be dependent from sensor to sensor.

$$T_k(\underline{y}_k) = \left. \frac{\frac{d^\ell}{d\theta^\ell} f_{Y_k}(\underline{y}_k | \theta)}{f_{Y_k}(\underline{y}_k | \theta)} \right|_{\theta=0} + \left. \frac{\sum_{\underline{u}^k} C(\underline{u}^k) \left[\frac{d^\ell}{d\theta^\ell} \text{Prob}(\underline{U}^k = \underline{u}^k | \underline{Y}_k = \underline{y}_k, \theta) \right]}{\sum_{\underline{u}^k} C(\underline{u}^k) [\text{Prob}(\underline{U}^k = \underline{u}^k | \underline{Y}_k = \underline{y}_k, \theta)]} \right|_{\theta=0} \quad (1)$$

The general model of (2) and (3) will allow us to discuss several interesting subcases. If both sensors observe a random signal with a nonzero mean then the LO sensor test statistics are of the same form as the LO test statistic used by an isolated sensor to detect a known signal in noise. The more interesting case is where both sensors observe a random signal with zero mean as we now illustrate. Assuming sufficient regularity of the noise pdf's, the LO sensor test statistic for the first sensor detector has been shown [24] to be of the form

$$T_1(\underline{y}_1) = \sum_{i=1}^m E\{S_{1i}^2\} \frac{f''_{W_1}(\underline{y}_{1i})}{f_{W_1}(\underline{y}_{1i})} + \sum_{i=1}^m \sum_{j=1, j \neq i}^m E\{S_{1i}S_{1j}\} \frac{f'_{W_1}(\underline{y}_{1i}) f'_{W_1}(\underline{y}_{1j})}{f_{W_1}(\underline{y}_{1i}) f_{W_1}(\underline{y}_{1j})} + \sum_{i=1}^m \sum_{j=1}^m a_{1j} E\{S_{1i}S_{2j}\} \frac{f'_{W_1}(\underline{y}_{1i})}{f_{W_1}(\underline{y}_{1i})} + b_1 \quad (4)$$

and T_1 is compared to the threshold λ_1 to make a decision. In (4)

$$a_{1j} = 2 \frac{c_\Delta}{\alpha_2 c_\Delta + c_b} \left[\int_{T_2 > \lambda_2} \frac{f'_{W_2}(\underline{y}_{2j})}{f_{W_2}(\underline{y}_{2j})} f_{W_2}(\underline{y}_2) d\underline{y}_2 \right] \quad (5)$$

and $b_1 = c_1 + d_1$ where (see (6) and (7) at the bottom of the page). Here we have used the symbol α_j to denote the false-alarm probability of the j th local sensor test for $j = 1, 2$ and T_2 and λ_2 denote, respectively, the test statistic and threshold used at sensor two. The constants c_Δ and c_b depend on the fusion rule $P(U_0 = u_0 | u_1, u_2) = \text{Prob}(U_0 = u_0 | U_1 = u_1, U_2 = u_2)$ and are given by

$$c_\Delta = P(U_0 = 1 | 1, 1) - P(U_0 = 1 | 0, 1) - [P(U_0 = 1 | 1, 0) - P(U_0 = 1 | 0, 0)] \quad (8)$$

and

$$c_b = P(U_0 = 1 | 1, 0) - P(U_0 = 1 | 0, 0). \quad (9)$$

The test statistic for the other sensor detector is similar to (4), with all references to sensor detector one and two interchanged.

The test statistic in (4) consists of four terms. The constant fourth term b_1 is not a function of $\underline{y}_1$ and can be considered to be part of the threshold of the test. The first two terms are familiar from centralized detection theory while the third term appears to be unique to distributed detection cases. The first two terms of (4) form a generalized energy measure [2] based on the observations at the first

sensor. These two terms of (4) can also be identified as the LO centralized test statistic for an isolated sensor. The third term in (4) is not found in the LO test for an isolated sensor, and so if this term is not identically zero it can make the distributed sensor test statistics quite different in functional form (as a function of $\underline{y}_1$) from the LO test for an isolated sensor. Further, it is possible that this part can be different at each sensor detector, making the LO test statistic at each sensor detector distinct.

While the physical interpretation of the third term in (4) may not be initially obvious, this part appears to be an attempt to form a correlation term. In fact, the entire test statistic in (4) can be interpreted as an approximation to the corresponding two-sensor LO centralized test statistic. Since the first sensor detector does not have the data needed to construct the two-sensor centralized LO test statistic, it attempts to approximate the desired correlation processing in the LO two-sensor centralized test. To illustrate these points we consider the LO centralized test statistic for the same problem which is

$$T_{\text{cen}}(\underline{y}_1, \underline{y}_2) = \sum_{i=1}^m E\{S_{1i}^2\} \frac{f''_{W_1}(\underline{y}_{1i})}{f_{W_1}(\underline{y}_{1i})} + \sum_{i=1}^m \sum_{j=1, j \neq i}^m E\{S_{1i}S_{1j}\} \frac{f'_{W_1}(\underline{y}_{1i}) f'_{W_1}(\underline{y}_{1j})}{f_{W_1}(\underline{y}_{1i}) f_{W_1}(\underline{y}_{1j})} + \sum_{i=1}^m \sum_{j=1}^m 2E\{S_{1i}S_{2j}\} \frac{f'_{W_1}(\underline{y}_{1i}) f'_{W_2}(\underline{y}_{2j})}{f_{W_1}(\underline{y}_{1i}) f_{W_2}(\underline{y}_{2j})} + \sum_{i=1}^m \sum_{j=1, j \neq i}^m E\{S_{2i}S_{2j}\} \frac{f'_{W_2}(\underline{y}_{2i}) f'_{W_2}(\underline{y}_{2j})}{f_{W_2}(\underline{y}_{2i}) f_{W_2}(\underline{y}_{2j})} + \sum_{i=1}^m E\{S_{2i}^2\} \frac{f''_{W_2}(\underline{y}_{2i})}{f_{W_2}(\underline{y}_{2i})}. \quad (10)$$

By comparing the LO centralized test statistic in (10) to the LO distributed test statistic at the first sensor in (4), we find that the distributed sensor detector does form those terms in (10) that it has the data to form. Specifically, the first two terms of (4) exactly match the first two terms of (10). On the other hand, the terms that the first sensor detector does not have all the data to form can only be approximated. The approximation uses a conditional expected value given that signal presence has been determined at the other sensor, which is reasonable since the unconditional expected value would be zero. The third term in (4) approximates the third term in (10). This term in (10) consists of quantities that each depend on observations from pairs of sensors. These

$$c_1 = \frac{c_\Delta}{\alpha_2 c_\Delta + c_b} \left[\sum_{i=1}^m E\{S_{2i}^2\} \int_{T_2 > \lambda_2} \frac{f''_{W_2}(\underline{y}_{2i})}{f_{W_2}(\underline{y}_{2i})} f_{W_2}(\underline{y}_2) d\underline{y}_2 \right] \quad (6)$$

$$d_1 = \frac{c_\Delta}{\alpha_2 c_\Delta + c_b} \left[\sum_{i=1}^m \sum_{j=1, j \neq i}^m E\{S_{2i}S_{2j}\} \times \int_{T_2 > \lambda_2} \frac{f'_{W_2}(\underline{y}_{2i}) f'_{W_2}(\underline{y}_{2j})}{f_{W_2}(\underline{y}_{2i}) f_{W_2}(\underline{y}_{2j})} f_{W_2}(\underline{y}_2) d\underline{y}_2 \right]. \quad (7)$$

Table 1 The Best Parameters for Sensor Detector 1 and Fusion Rule for Mixture Noise. ($a_{21} = a_{11}$ for AND Rule Optimum and $a_{21} = -a_{11}$ for the OR Rule Optimum. $b_1 - \tau = b_2 - \tau$ in All Cases).

α_0	$\epsilon = 0.1$			$\epsilon = 0.2$			$\epsilon = 0.3$		
	Fusion	a_{11}	$b_1 - \tau$	Fusion	a_{11}	$b_1 - \tau$	Fusion	a_{11}	$b_1 - \tau$
0.01	AND	0.0	-2.05365	AND	0.0	-1.990898	AND	2.924756	-5.011461
0.1	OR	0.0	-2.849148	AND	0.0	-0.410542	AND	0.0	-0.654041
0.2	OR	0.0	-1.955859	OR	0.0	-1.945252	AND	0.0	+0.030801
0.3	OR	0.0	-1.35687	OR	0.0	-1.511671	OR	0.337433	-1.402628

quantities give a generalized measure of the correlation between the observations at the two sensors. In the case of Gaussian noise

$$-\frac{f'_{W_k}(y_{ki})}{f_{W_k}(y_{ki})} = y_{ki}$$

and the third term of (10) becomes what is normally called a correlation operation. Similarly, the constant term b_1 in (4) approximates the last two terms in (10). In this case none of the data needed to calculate the last two terms in (10) is available at sensor one.

The numerical solutions in [24] indicate how the optimum sensor and fusion rules may vary with noise pdf, false-alarm probability, and m . The conditions specified by (4)–(7), their counterparts at the second sensor, and the false-alarm constraint represent a set of necessary conditions for an LO distributed test for a given fusion rule. In [24], the solutions to these necessary conditions are searched for and the best sensor and fusion rules found are studied for some common-signal cases ($S_{1i} = S_{2i}, i = 1, \dots, m$) with specific noise pdf's. These results focus on two-sensor cases that use AND and OR fusion rules, $\lambda_1 = \lambda_2 = \lambda$ and several specific false-alarm probabilities α_0 between 0.01 and 0.3. We now highlight these results.

Initially consider the single observation per sensor ($m = 1$) case. For the unit-variance zero-mean Gaussian noise cases, the best choice of a_{11} and a_{21} is $a_{11} = a_{21} = 0$, the optimum isolated sensor test, and for this case the OR rule gives better performance. On the other hand, for heavy tailed noise pdfs nonzero a_{11} and a_{21} are sometimes found to give best performance. For Cauchy noise, with pdf $f_{W_k}(y) = 1/[\pi(1+y^2)]$, the best solutions using AND fusion rules outperform the best OR rule solutions. In these cases a_{11} is nonzero for best performance for small α_0 , but for larger α_0 the best choice is $a_{11} = 0$. In each case $a_{11} = a_{21}$ for best performance. Similar results are found for a noise pdf that is a mixture of two Gaussian pdfs, one with unit variance and the other with a larger variance σ^2 . In this case the pdf of the noise is

$$f_{W_k}(y) = (1 - \epsilon) \frac{1}{\sqrt{2\pi}} e^{-\frac{y^2}{2}} + \epsilon \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{y^2}{2\sigma^2}}. \quad (11)$$

Table 1 describes the best processing at the first sensor for three different instances of the above model ($\epsilon = 0.1, 0.2$, and 0.3). All the cases have $\sigma^2 = 10$. In these cases the best fusion rule is an AND rule for smaller α_0 but switches to an OR fusion rule for larger α_0 . For the $\epsilon = 0.1$ and the $\epsilon = 0.2$ cases the best choice for a_{11} and a_{21} is zero, but for $\epsilon = 0.3$ the best a_{11} and a_{21} are nonzero for small α_0 and for α_0 in a region around $\alpha_0 = 0.3$. In the region

near $\alpha_0 = 0.3$ the OR rule is better and the best sensor tests use $a_{21} = -a_{11}$. For small α_0 the AND rule is best and $a_{11} = a_{21}$.

One interesting observation that can be made from the above results is that whenever the AND fusion rule is optimum it is found that $a_{11} = a_{21}$ is best. When an OR rule is optimum $a_{11} = -a_{21}$ is best. In fact it has been shown analytically that for best performance a_{11} and a_{21} must have the same signs if an AND rule is used and that they must have different signs if an OR rule is used. This is a consequence of the sensor detectors trying to work together to implement a specific strategy. If the AND rule is optimum each sensor essentially tests for a signal of the same sign, a form of correlation detection, while if the OR rule is optimum each sensor detector will essentially test for a signal of a different sign, a form of energy detection. This idea is illustrated in Fig. 1 which shows the LO test statistics T_1 and T_2 described by (4) for the case of $\alpha_0 = 0.1$ with the best parameter values found for a specific case with Cauchy noise. Both sensor tests are identical and each sensor responds to a set of positive observations that signify a positive signal.¹ In the case where the OR rule is optimum, which is illustrated in Fig. 2, one sensor responds to a set of positive observations, while the other sensor responds to a set of negative observations.

Similar results are obtained for cases with $m > 1$. The most noteworthy property of the solutions for these cases is that the a_{kj} coefficients tend to decrease as m increases. This indicates that the optimum sensor test statistics approach the optimum isolated sensor test statistics as $m \rightarrow \infty$. This is verified in [24] by an asymptotic analysis.

B. Narrowband Signals

In communications, radar, and sonar problems narrowband signals are frequently of interest. In these cases the models in (2) and (3) can be used to represent the observations using in-phase (I) and quadrature (Q) notation so that $Y_{ki} = (Y_{Iki}, Y_{Qki})$, $S_{ki} = (S_{Iki}, S_{Qki})$, and $W_{ki} = (W_{Iki}, W_{Qki})$ for $k = 1, 2$ and $i = 1, \dots, m$ [25]. For simplicity, assume that $\{W_{1i}, W_{2i}, i = 1, \dots, m\}$ is a set of independent bivariate zero-mean noise samples, each with the common bivariate pdf f_{IQ} which is circularly symmetric so that $f_{IQ}(u, v) = h(\sqrt{u^2 + v^2})$ for a known function h . Further assume that the signal samples at different time instances, S_{ki} and $S_{\ell j}, i \neq j$, are independent but signal samples at the same time instance, S_{1i} and S_{2i} , are dependent and $E\{S_{1i}S_{2i}\} = E\{S_{ki}^2\} = 1, k = 1, 2$.

¹ A dual solution where each sensor detector responds to negative signals is also optimum.

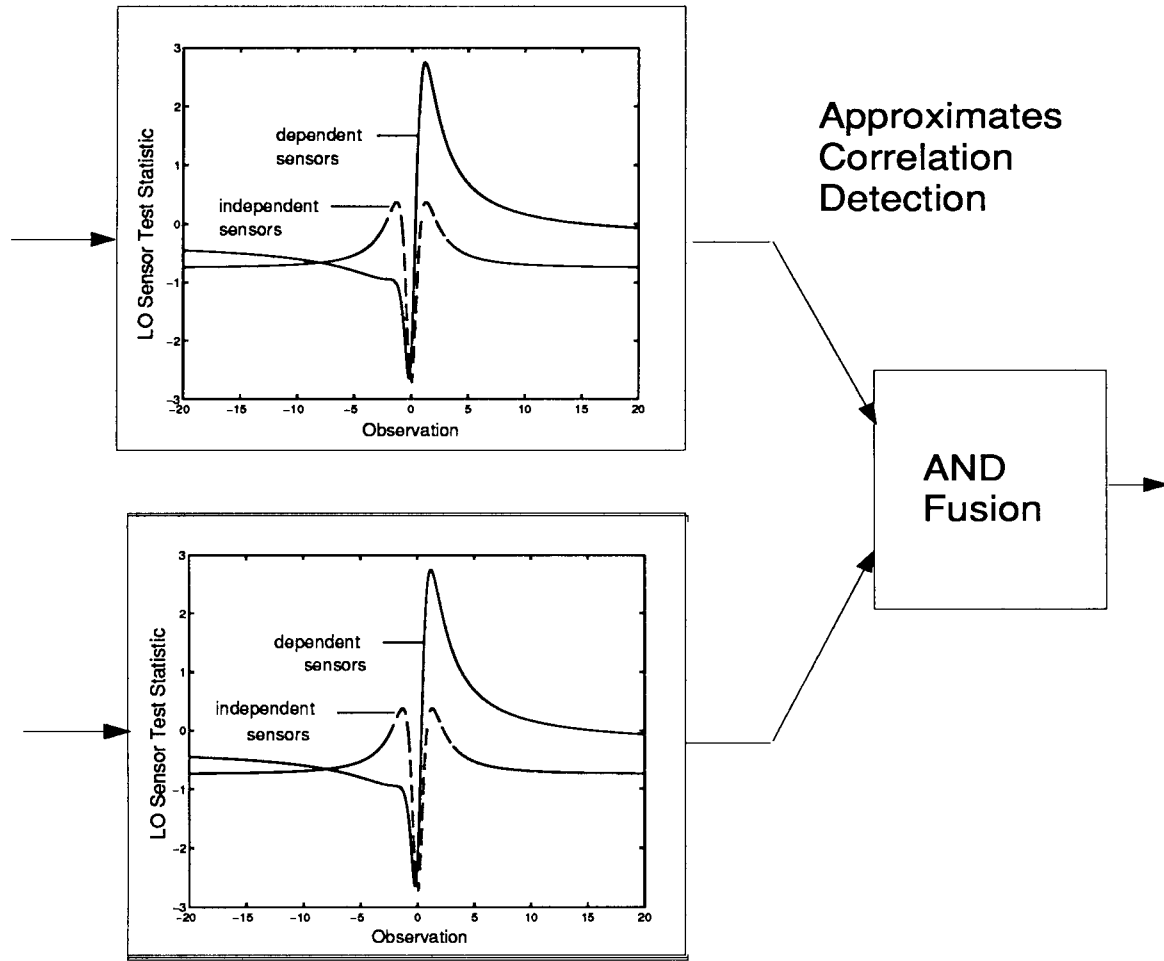


Fig. 1. Interpretation of distributed scheme with AND fusion.

Under these assumptions the optimum test statistic at the first sensor is

$$T_1(\underline{y}_1) = \sum_{i=1}^m \left[\frac{h''(r_{1i})}{h(r_{1i})} + \frac{h'(r_{1i})}{r_{1i}h(r_{1i})} + (a_{I1i}y_{I1i} + a_{Q1i}y_{Q1i})g_{LO}(r_{1i}) \right] + b_1 \quad (12)$$

with

$$\gamma_{ki} = \sqrt{Y_{Iki}^2 + Y_{Qki}^2}$$

and

$$a_{I1i} = 2 \frac{c_\Delta}{\alpha_2 c_\Delta + c_b} \int_{T_2(\underline{y}_2) > \lambda_2} y_{I2i} g_{LO}(r_{2i}) \times \prod_{j=1}^m f_{IQ}(y_{I2j}, y_{Q2j}) dy_{I2j} dy_{Q2j} \quad (13)$$

$$a_{Q1i} = 2 \frac{c_\Delta}{\alpha_2 c_\Delta + c_b} \int_{T_2(\underline{y}_2) > \lambda_2} y_{Q2i} g_{LO}(r_{2i}) \times \prod_{j=1}^m f_{IQ}(y_{I2j}, y_{Q2j}) dy_{I2j} dy_{Q2j} \quad (14)$$

$$b_1 = \frac{c_\Delta}{\alpha_2 c_\Delta + c_b} \int_{T_2(\underline{y}_2) > \lambda_2} \sum_{i=1}^m \left(\frac{h''(r_{2i})}{h(r_{2i})} + \frac{h'(r_{2i})}{r_{2i}h(r_{2i})} \right)$$

$$\times \prod_{j=1}^m f_{IQ}(y_{I2j}, y_{Q2j}) dy_{I2j} dy_{Q2j} \quad (15)$$

with $g_{LO}(r) = -h'(r)/(rh(r))$ and where $h'(r)$ and $h''(r)$ denote the first and second derivatives of the function h with respect to the amplitude argument r . The test statistic for the second sensor is very similar to (12), with all references to sensor one and two interchanged. The first two terms in (12) again correspond to the optimum test for an isolated sensor, while the third term again forms an estimate of the correlation measure formed in the corresponding centralized test [25]. Numerical results in [25] show a similar symmetry between the (a_{I1i}, a_{Q1i}) and the (a_{I2i}, a_{Q2i}) coefficients that again seems to result from the sensor decision and fusion rules working together to approximate energy and correlation detection schemes. Further, the size of $a_{I1i}^2 + a_{Q1i}^2$ and $a_{I2i}^2 + a_{Q2i}^2$ for $i = 1, \dots, m$ tend to decrease as m gets large.

III. NONPARAMETRIC AND CFAR DISTRIBUTED DETECTION

There has been extensive research on nonparametric detection techniques for centralized signal detection problems, based on nonparametric hypothesis testing formulations [3],

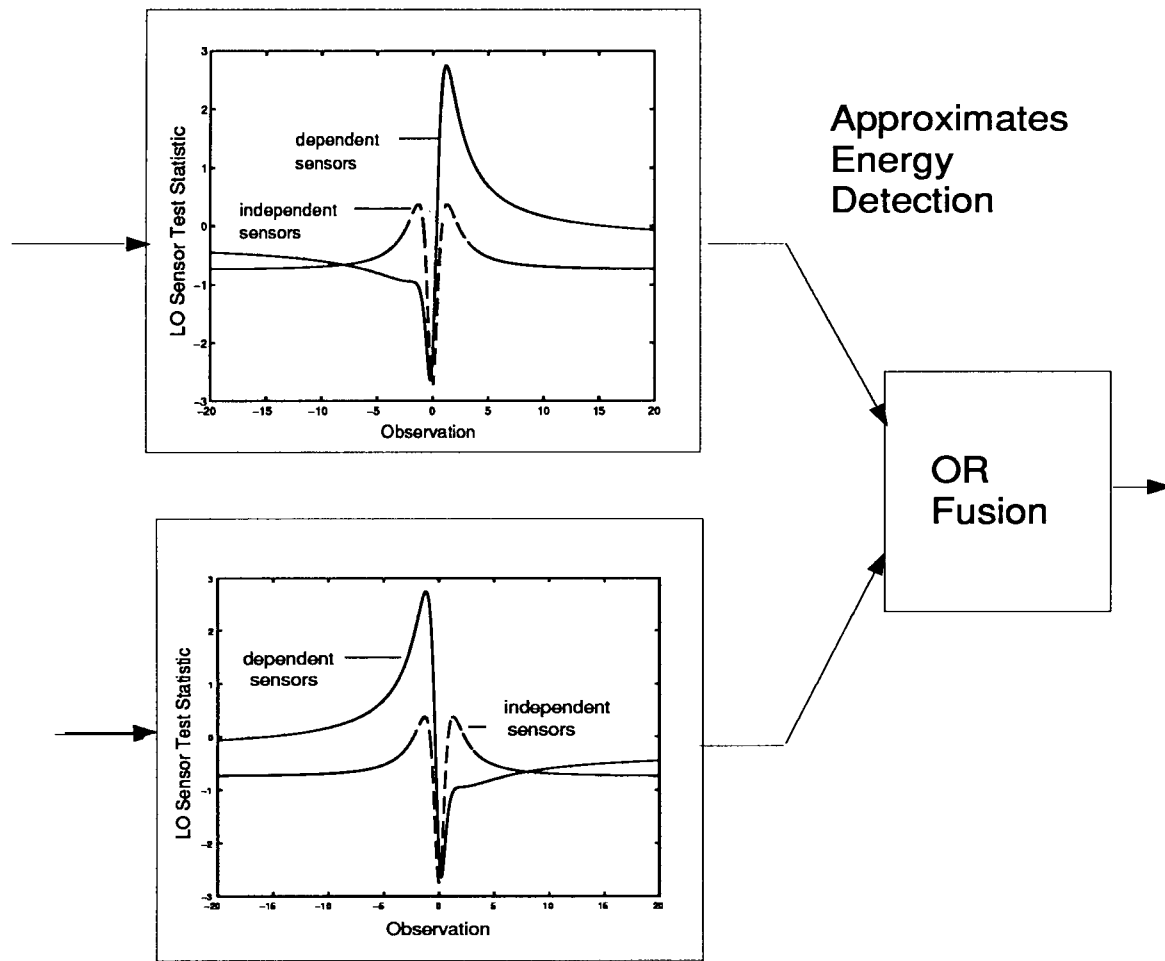


Fig. 2. Interpretation of distributed scheme with OR fusion.

[5], [10] (see [4] for a recent survey). A nonparametric hypothesis does not specify the detailed functional forms of the joint distributions; the joint distributions under a nonparametric hypothesis form a large class of possibilities that can not be indexed by a finite number of parameters. Many of the resulting centralized tests are based on signs or rank orderings of the observations. In the decentralized setting, nonparametric detection techniques have only recently become of interest.

Considerable attention has also been focused on CFAR detection schemes. Generally, these schemes assume parametric models for their observations, but they are often classified as nonparametric schemes. For signals in nonstationary and possibly unknown environments, CFAR detectors have been found to be useful. In radar and sonar applications for example, signal returns must be distinguished from a background consisting of thermal noise and clutter, which refers to any undesired signal reflections from objects other than the target. Typically the clutter-plus-noise power is unknown at the various points in space probed for a target. CFAR techniques determine a new threshold setting for each point probed with the goal of maintaining a constant false-alarm probability. The threshold is set using a set of reference observations, which are typically obtained from probing near-by points in space.

A. Nonparametric Distributed Detection

Nonparametric schemes that are based on signs and ranks can be quite good in their detection performance under nominal conditions of Gaussian noise, and their performance remains relatively good in other cases also. In many detection scenarios, they provide a constant false-alarm probability that is independent of the exact noise distribution as long as the distribution is symmetric. In some applications, information may be available as to specific statistical descriptions that are highly likely to be encountered. Such information is usually stated in terms of a nominal statistical model and it can be used to design good nonparametric detection schemes.

A nonparametric decentralized detection scheme was studied in [26]. In [26], Wilcoxon signed-rank test statistics are sent with full precision to the fusion center. Cases with binary sensor decisions are also discussed. The use of Wilcoxon statistics to generate multi-bit sensor decisions is considered in [27]. The results in [26] indicate that the Wilcoxon-based decentralized detection scheme considered performs well for cases with heavy tailed non-Gaussian noise when compared with schemes designed for Gaussian noise cases when the comparison uses an asymptotic performance measure. In [28], the sign detector

and the dead-zone detector are studied in a decentralized context. Some generalizations of sign and dead-zone detectors have also been studied. The conditional sign test and the conditional dead-zone limiter tests are studied in [29] for distributed detection. In [30], a distributed scheme which uses generalized sign tests to generate binary decisions at each sensor is proposed. The generalized sign test first subtracts a reference observation from each sensor observation and then counts the number of these subtractions which result in a positive difference.

The Wilcoxon signed-rank test statistic, sign detector, dead-zone detector, and their extensions have received extensive attention for centralized cases. Each of these tests provides good performance over a fairly wide range of operating conditions, but none is optimum in terms of detection performance for any particular operating environment that is of interest. Nonparametric schemes based on signs and ranks can be constructed that provide LO performance under a given nominal model for centralized signal detection problems. For example, in [31] the focus is on weak random signals and performance is optimized for nominal noise distributions. The results in [31] were recently extended for distributed cases in [32].

Consider the detection of a weak zero-mean random signal observed in additive zero-mean noise as described by (2) and (3) for the case of the common noise pdf $f_{W_1} = f_{W_2} = f$. In [32], the optimum sensor tests based on the signs and ranks of the observations are derived for weak signals and known f and a methodology is described to optimize the design of the overall distributed detection scheme to obtain best performance for a known f . For cases with unknown f , [32] considers using the scheme that results when f is taken to be the nominal model.

For simplicity consider the case of identical signals at each sensor $S_{1i} = S_{2i}$ and where f is assumed to be a generalized Gaussian pdf as given by

$$f(x) = \frac{\nu}{2A(\nu)\Gamma(1/\nu)} \exp\left[-\left(\frac{|x|}{A(\nu)}\right)^\nu\right] \quad (16)$$

where $\Gamma(a) = \int_{x=0}^{\infty} x^{a-1} \exp(-x) dx$ is the Gamma function and

$$A(\nu) = \sqrt{\frac{\Gamma(1/\nu)}{\Gamma(3/\nu)}}. \quad (17)$$

If $\nu = 2$ then f is the Gaussian pdf, and if $\nu < 2$ then f is a pdf with heavy tails. The heavy-tailed pdfs appear to be reasonable models for a number of practical cases [2] including the case of impulsive noise.

Consider the case where the actual noise has a generalized Gaussian pdf with a parameter ν that is different from the assumed value of ν_d used for design. Now compare the variation in false-alarm probability of the LO distributed scheme in (4) to the variation in false-alarm probability of a scheme in which the sensor tests are based only on the signs and ranks of the sensor observations. Since the probability mass function of the signs and ranks is the same for any symmetric f , it is easy to show that there is no variation in

the false-alarm probability for the schemes based on signs and ranks for any actual value of ν . On the other hand, the scheme in (4) generally exhibits a false-alarm probability that varies with the actual value of ν and the variation can be large in some cases.

Next consider the probability of detection $P_d(\theta)$ of each of the two schemes under a similar set of conditions. To be specific, consider the case where the two schemes are designed for $\nu_d = 2$, the case of Gaussian noise. Further, to focus on the sensor test, consider the case where each scheme uses an AND fusion rule. If the actual noise has $\nu = 2$, Monte Carlo simulation results presented in [32] indicate that the performance of the two schemes is comparable for small- θ cases. On the other hand, if the true ν represents a sufficiently heavy tailed pdf, then the scheme based on signs and ranks is much better [32]. An intuitive explanation is that the large observations, which are much more probable with heavy-tailed pdfs, tend to dominate the test in (4), which leads to poor performance. The schemes based on ranks are not influenced by the actual magnitude of the observations and so they are not influenced as much by heavy-tailed pdf's.

B. CFAR Distributed Signal Detection

Several techniques have been proposed for setting the threshold in centralized CFAR signal detection systems [33] and many of these techniques have also been investigated for distributed cases [34], [35]. The first investigation [34] considered a cell-averaging (CA) CFAR scheme, where the thresholds are proportional to the average of the square envelope of the reference observations. Denote the square envelope of the matched-filtered single-pulse return for the cell-under-test at the k th sensor as Y_k and denote the set of L noise-plus-clutter-only square envelope reference observations at the k th sensor as $X_{k1}, \dots, X_{kL}$. In [34] the k th sensor test compares Y_k to $t_k \sum_{q=1}^L X_{kq}$ where t_k is called the *fixed part* of the threshold. Further, the fixed parts of the sensor thresholds are chosen to simultaneously provide the required false-alarm probability and to maximize detection probability for a case with a known signal-to-noise-plus-clutter ratio and Gaussian noise-plus-clutter complex envelope observations that are independent from sensor to sensor. The results in [34] are for the parallel distributed sensor topology [1], and similar results for a serial distributed sensor topology appear in [35]. The probability of detection performance obtained when using a particular nonparametric scheme at each sensor is compared to the performance obtained when using a CA-CFAR scheme at each sensor in [30]. For the particular case considered in [30] CA-CFAR performs very well.

The CA-CFAR scheme appears to perform poorly for cases where clutter varies rapidly or for detection in the presence of interferers [33], but a scheme in which the order statistics of the reference observations are used to compute the threshold, called order statistic constant false-alarm rate (OS-CFAR) detection, appears promising. There have recently been investigations of OS-CFAR for some distributed sensor cases in [36]–[38]. Optimum design

procedures are considered for a parallel topology case with a known signal-to-noise-plus-clutter ratio and Gaussian noise-plus-clutter complex envelope observations which are independent from sensor to sensor in [36]–[38]. Analysis of cases with nonhomogeneous clutter and interfering targets is provided in [37]. Locally optimum detection techniques, similar to those discussed previously, have been used to design optimum CA [39] and OS [40] CFAR schemes for detecting a weak random signal in Gaussian noise-plus-clutter of unknown and possibly varying power. In [39] and [40], cases are considered where the observations are statistically dependent from sensor to sensor. The results in [37], [39], [40] indicate that OS schemes appear to provide better performance for cases with nonhomogeneous clutter and that the choice of a fusion rule may involve trading off detection performance in a homogeneous and non-homogeneous environment. Locally optimum distributed detection concepts are used to design CFAR distributed detection schemes in [41] for cases with multiple bit sensor decisions and observations that are independent from sensor to sensor. Locally optimum schemes are considered in [42] for an interesting class of non-Gaussian noise-plus-clutter distributions that have unknown scale parameters. Further research on distributed CFAR signal detection is reported in [43] through [45].

IV. ROBUST DISTRIBUTED DETECTION

The robust detection design philosophy is closely related to the nonparametric detection design philosophy. We discuss briefly the issue of robustness of distributed detection systems to uncertainties in the statistical models used for algorithm design. In the context of centralized detection, this problem of robustness has been considered in some depth (see [6] and [7] for surveys).

A basic principle of robust centralized detection is that robust detectors can be designed by designing *optimum* detectors for certain *least favorable* models. As an example, consider the centralized detection problem of deciding between two possible marginal distributions (P_0 and P_1) for a stream $Y_1, Y_2, \dots, Y_k$ of i.i.d. observations. There are several optimality criteria of interest for detector design [3]. These include:

- 1) *Bayesian detection*, which minimizes the average probability of error (i.e., the average of the false-alarm probability and the miss probability, weighted by prior probabilities on the two distributions);
- 2) *minimax detection*, which minimizes the maximum of the false alarm and miss probabilities; and
- 3) *Neyman–Pearson detection*, which minimizes the miss probability within an upper bound constraint on the false-alarm probability.

Each of these problems is solved by a likelihood ratio test, which decides between P_0 and P_1 on the basis of the value of the likelihood ratio

$$\prod_{i=1}^k L(Y_i) \quad (18)$$

relative to a threshold. Here $L(Y_i)$ is the likelihood ratio between P_0 and P_1 based on the single sample Y_i . The value of the threshold is determined by which of the above three criteria is of interest. Note that, if P_j has pdf p_j for $j = 0, 1$, then

$$L(Y_i) = \frac{p_1(Y_i)}{p_0(Y_i)}, \quad i = 1, 2, \dots \quad (19)$$

A useful way of seeking robustness in this framework is to replace the simple hypotheses that the marginal distribution of the observations is either P_0 or P_1 , with composite hypotheses that the marginal distribution of the observations is drawn from one of two *uncertainty classes* $\mathcal{P}_0$ and $\mathcal{P}_1$ of marginal distributions. The idea is that $\mathcal{P}_0$ represents a range of possible statistical behavior under the hypothesis H_0 , and similarly for $\mathcal{P}_1$. For example, $\mathcal{P}_0$ might be the set of all distributions of the form

$$(1 - \epsilon)P_0 + \epsilon Q \quad (20)$$

where P_0 is a *nominal* marginal distribution for the observations, $\epsilon \in [0, 1]$ is a degree of uncertainty that the designer places in the nominal model, and Q is an arbitrary and unknown *contaminating* distribution. Such a class is called an ϵ -mixture class. Similarly, $\mathcal{P}_1$ could be an ϵ -mixture class around a nominal distribution P_1 . There are, of course, a number of other possible ways in which the uncertainty classes $\mathcal{P}_0$ and $\mathcal{P}_1$ can be defined.

Since the criteria 1)–3) cannot be achieved uniformly over general uncertainty classes $\mathcal{P}_0$ and $\mathcal{P}_1$, these problems can be reformulated within this context of uncertainty by replacing the false-alarm and miss probabilities with their maximum values over the sets $\mathcal{P}_0$ and $\mathcal{P}_1$, respectively. Detectors that optimize these new criteria are known as *minimax robust detectors*. For many types of uncertainty classes of interest, such detectors can be designed by first finding a pair of least-favorable distributions in $\mathcal{P}_0$ and $\mathcal{P}_1$, and then designing an optimum (likelihood-ratio) test between these least-favorable distributions.

For example, when the uncertainty classes are ϵ -mixture classes, the resulting test compares the quantity

$$\prod_{i=1}^k [L(Y_i)]_a^b \quad (21)$$

to a threshold, where $L(Y_i)$ is the single-observation likelihood ratio for the nominal model; $0 < a < b < \infty$ are two constants depending on the nominal model and the value of ϵ ; and the notation $[x]_a^b$ means

$$[x]_a^b = \begin{cases} b & \text{if } x \geq b \\ x & \text{if } a < x < b \\ a & \text{if } x \leq a. \end{cases} \quad (22)$$

Note that this detector robustifies against uncertainty by limiting the value of the nominal likelihood ratio in regions of greatest sensitivity—i.e., when it is either large or close to zero. In this case, the least-favorable distributions are essentially the elements of $\mathcal{P}_0$ and $\mathcal{P}_1$ that look as much like one another as possible. Similar robust detectors and least-favorable distributions are obtained with other uncertainty classes [6], [7].

The issue of robustness has been considered in the context of distributed detection in several works, including [46]–[48]. In [47] minimax robustness is considered for distributed detection networks both with and without fusion centers [1, Figs. 1 and 2]. In each case, within mild assumptions, the above-noted basic principle holds—namely, *minimax robustness is achieved in distributed detection by designing optimum distributed detectors for least-favorable models*. Moreover, again within mild assumptions, these least-favorable models are essentially the same for distributed detection as they are for centralized detection. Thus, the robust distributed detection problem reduces to the design of optimal distributed detection systems [1, [23] for least-favorable models. So, for example, if the observation distributions in a distributed detection network are uncertain, and this uncertainty can be modeled with ϵ -mixture models, then robustness can be achieved by limiting the various likelihood ratios as in (21). Of course, the degree of limiting (represented by a and b) will depend on the values of ϵ placed on the various distributions. An asymptotic version of this problem is considered in [46], and robustness with respect to uncertainty in a parametric prior is considered in [48].

V. SEQUENTIAL METHODS IN DISTRIBUTED DETECTION

The distributed detection procedures described in the preceding sections and in Part I of this paper are *fixed sample size* detectors; that is, they each operate with a fixed number of observations that is predetermined at the time the detector is designed. An alternative type of detection procedure is a *sequential* detector, in which the number of observations used in deciding between hypotheses is a random quantity that depends on the observations themselves. In such procedures, there is usually a large (possibly infinite) number of observations that can be taken in sequence, and the decision as to when to stop taking observations is part of the overall detection procedure. The performance indices used for designing such detectors include costs for taking observations, so that a tradeoff emerges between detection quality (measured by error probabilities) and average sampling costs.

The issue of sequential processing has been considered within the framework of distributed detection. In this section, we review briefly some recent results in this area. There are three basic situations that have been considered in this context. The first situation is that in which individual sensors in a distributed detection network pass intermediate decisions to a fusion center, which then performs a sequential test on these intermediate decisions [1, Fig. 1]. The second situation is that in which the individual sensors themselves perform sequential tests, and they behave as a team due to a coupled objective function [1, Fig. 2]. The third situation involves the problem of *quickest detection*, in which the detection of an abrupt change at an unknown time is desired. In this case, topologies both with and without fusion centers have been considered.

Before describing results in these areas, it is useful to consider briefly the problem of centralized sequential detection.

A. Centralized Sequential Detection

Centralized sequential detection is a well-developed area [3]. The design of sequential detectors involves the choice of two procedures: a *stopping rule* and a *terminal decision rule*. The stopping rule is a function that decides, on the basis of sequential observation, when to stop taking observations. The *stopping time* is the time at which the stopping rule decides to stop taking observations; the stopping time is a random variable since it depends on the observations. The terminal decision rule is a decision rule that is applied to the observations that have been taken when the stopping rule stops.

To illustrate this idea, consider the problem of testing between two statistical hypotheses H_0 and H_1 concerning an infinite sequence of i.i.d. random observations $Y_1, Y_2, \dots$. Let us denote by ν the stopping time of a sequential test, and by δ_ν the terminal decision rule. Assuming that prior probabilities can be assigned to the two hypotheses, an overall cost of testing can be written as

$$P_e(\delta_\nu) + cE\{\nu\} \quad (23)$$

where $P_e(\delta_\nu)$ is the average probability with which δ_ν makes errors, $E\{\nu\}$ is the average number of observations used by the detector, and $c > 0$ is a per-sample cost of taking observations (or of decision delay). This performance index is minimized [3] by the stopping time

$$\nu = \min\{i \geq 1 \mid \pi_i \notin (\pi_L, \pi_U)\} \quad (24)$$

where

$$\pi_i = P(H_1 \mid Y_1, \dots, Y_i), \quad i = 1, 2, \dots \quad (25)$$

and where $0 \leq \pi_L < \pi_U \leq 1$ are two thresholds determined by the distributions of the observations. By (25) we mean the conditional probability that H_1 is true, given the observations $Y_1, \dots, Y_i$. The terminal decision rule chooses H_1 if $\pi_\nu \geq \pi_U$, and chooses H_0 otherwise (i.e., if $\pi_\nu \leq \pi_L$). The implementation of this test is facilitated greatly by the recursive relationship (which follows from Bayes formula)

$$\pi_i = \frac{L_i \pi_{i-1}}{L_i \pi_{i-1} + (1 - \pi_{i-1})}, \quad i = 1, 2, \dots \quad (26)$$

where $L_i = L(Y_i)$ is the likelihood ratio between H_0 and H_1 based on the i th observation, and where the initial condition π_0 is equal to the prior probability of H_1 .

The sequential test is illustrated in Fig. 3. The basic idea behind this test is clear—the test continues taking observations until sufficient evidence for one of the two hypotheses accumulates. It then stops and makes a decision. By allowing the flexibility to stop early when a decision is clear or to continue observing when the decision is unclear, an optimal trade-off between the cost of observations and the quality of decisions can be achieved. In the following subsections, we discuss results that extend this general idea to a distributed detection setting.

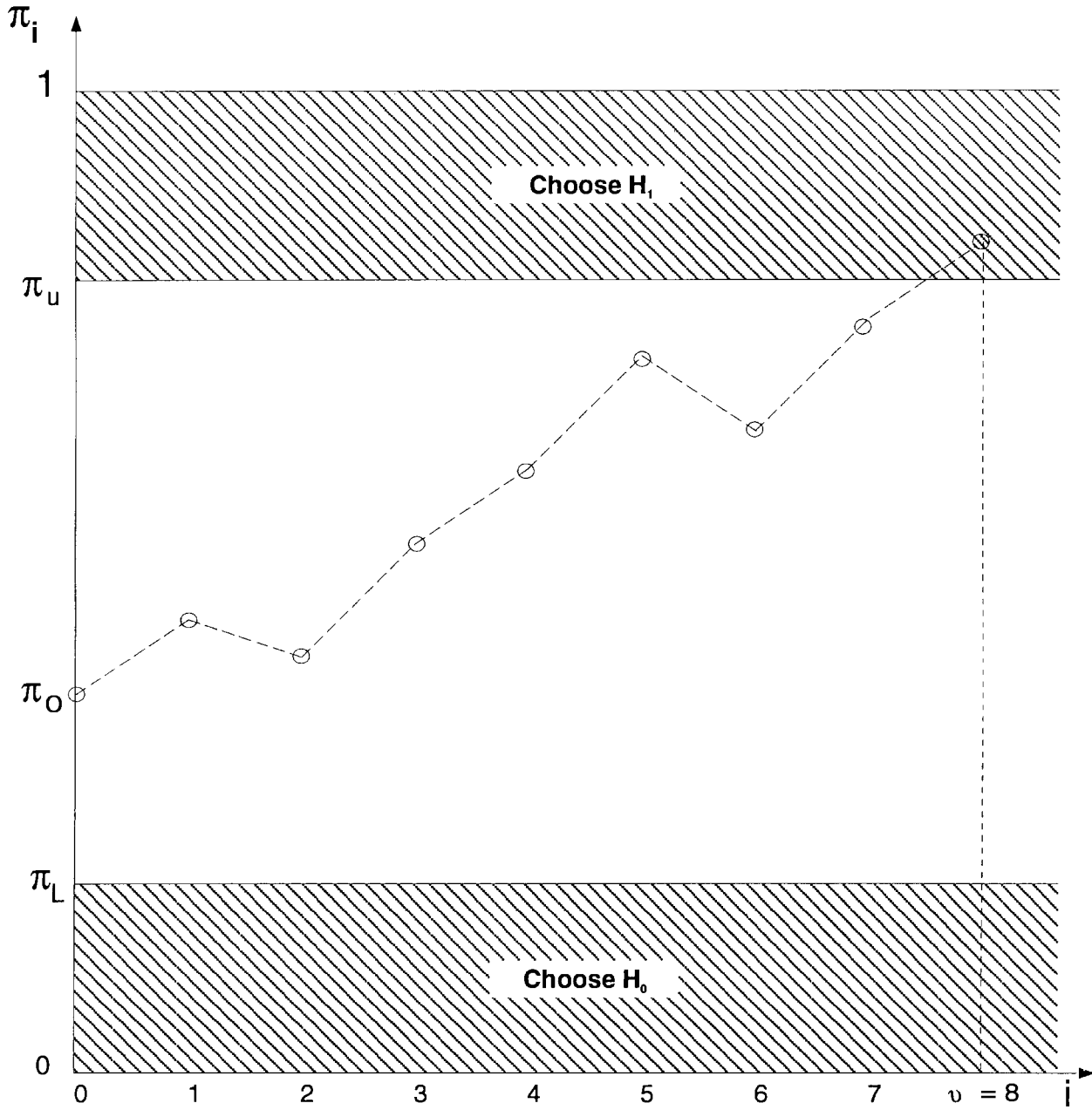


Fig. 3. Illustration of a Bayes sequential test.

B. Distributed Detection with Sequential Testing at the Fusion Center

The first situation we consider is that in which individual sensors in a distributed detection network pass intermediate decisions to a fusion center, which then performs a sequential test on these intermediate decisions. That is, the topology is that of [1, Fig. 1]. This problem has been considered in several works, including [49]–[52], [59]. Among the most recent and complete results in this area are those in [52], which we summarize here. Consider a network of N sensors, with observations $Y_{n1}, Y_{n2}, \dots$, at the n th sensor. It is assumed that these observations are independent from sensor to sensor, and that they are i.i.d. at each sensor. There are two statistical hypotheses, H_0 and H_1 , each of which corresponds to a distinct distribution for

the observations. At each time i the n th sensor computes a summary message u_{ni} , which is broadcast to all other sensors and to a fusion center. The summary message u_{ni} is a function of the observation Y_{ni} and of the past summary messages of all of the sensors; i.e.,

$$u_{ni} = \phi_{ni}(Y_{ni}; I_{i-1}) \quad (27)$$

where

$$I_{i-1} = u_{11}, \dots, u_{1(i-1)}, u_{21}, \dots, u_{2(i-1)}, \\ u_{N1}, \dots, u_{N(i-1)}, \quad i = 2, 3, \dots \quad (28)$$

and where ϕ_{ni} is a function of the given arguments. The idea is that these summary messages will be taken from a small alphabet (perhaps binary) that can be easily transmitted over a limited-capacity channel. In [52] it is

assumed that these alphabets are finite. For the purposes of discussion, we will assume here that these messages are binary; i.e., $u_{ni} \in \{0, 1\}$.

The fusion center is to perform a sequential test based on the summary messages from the sensors. That is, the fusion center will observe the summary messages, and then will decide when to stop observing and make a decision. The objective of the detector design is to choose the fusion center's stopping rule and decision rule, and the local sensors' message computation, in order to minimize a cost function comprised of a (linear) cost of sampling, plus an average probability of error as in (23).

The solution to this problem can be described in terms of the sequence of posterior probabilities

$$\tilde{\pi}_i = P(H_1 | I_i), \quad i = 1, 2, \dots \quad (29)$$

by which is meant [analogously with (25)] the conditional probability that the hypothesis H_1 is true, conditioned on the summary messages of all sensors up to and including time i . In particular, the fusion center performs a test similar to that for the centralized case (Fig. 3)—it uses the stopping time

$$\tilde{\nu} = \min\{i \geq 1 | \tilde{\pi}_i \notin (\tilde{\pi}_L, \tilde{\pi}_U)\} \quad (30)$$

where $0 \leq \tilde{\pi}_L < \tilde{\pi}_U \leq 1$ are again two thresholds determined by the distributions of the observations. The terminal decision rule chooses H_1 if $\tilde{\pi}_{\tilde{\nu}} \geq \tilde{\pi}_U$ and chooses H_0 otherwise.

Moreover, the summary messages broadcast by the distributed sensors are of the form

$$u_{ni} = \phi_n^*(Y_{ni}, \tilde{\pi}_{i-1}), \quad i = 1, 2, \dots \quad (31)$$

where the function $\phi_n^*(Y_{ni}, \tilde{\pi}_{i-1})$ takes on the value one if the local likelihood ratio between H_0 and H_1 based on Y_{ni} exceeds a threshold (depending on $\tilde{\pi}_{i-1}$), and it takes on the value zero otherwise. These thresholds used by the sensors are fixed functions of $\tilde{\pi}_{i-1}$ that are computed by solving an optimization problem described in [52].

From (30) and (31) we see that $\tilde{\pi}_{i-1}$ forms a sufficient statistic for the past information I_{i-1} , which greatly simplifies the structure of the test. Like π_i of (25) there is a recursive computation for the sequence $\tilde{\pi}_i$. This recursion is somewhat more complicated than (26) because it will depend on the functions used to compute the summary messages. In particular, for a given set of such functions, we have the recursion

$$\tilde{\pi}_i = \frac{\tilde{L}_i \tilde{\pi}_{i-1}}{\tilde{L}_i \tilde{\pi}_{i-1} + (1 - \tilde{\pi}_{i-1})}, \quad i = 1, 2, \dots \quad (32)$$

with $\tilde{\pi}_0$ equal to the prior probability of H_1 , and where $\tilde{L}_i$ is given by

$$\tilde{L}_i = \prod_{n=1}^N L^{(n)}(u_{ni} | \tilde{\pi}_{i-1}), \quad i = 1, 2, \dots \quad (33)$$

with $L^{(n)}(u_{ni} | \tilde{\pi}_{i-1})$ equal to the conditional likelihood ratio between H_0 and H_1 , based on the measurements

$u_{ni}, \dots, u_{Ni}$, and conditioned on $\tilde{\pi}_{i-1}$. Note that the function $L^{(n)}$ depends on the function ϕ_n^* of (31) via the relationship

$$L^{(n)}(u | \pi) = \frac{P_{n1}(\phi_n^*(Y_{n1}, \pi) = u)}{P_{n0}(\phi_n^*(Y_{n1}, \pi) = u)}, \quad u \in \{0, 1\}, \quad \pi \in [0, 1] \quad (34)$$

where P_{nj} denotes the marginal probability distribution of the Y_{ni} 's under hypothesis H_j .

These results specify an optimal distributed sequential procedure. It is intuitively pleasing in that each sensor computes its summary messages based on likelihood ratio comparisons, and the fusion center performs a test essentially like the centralized sequential test described in Section V-A. There are important distinctions, however, between this case and the centralized version. First, as with most distributed detection problems, the threshold computation for the local sensors is rather complicated. Secondly, the observations on the basis of which the fusion center is making its decision are not i.i.d. (Although those at the local sensors are.) Nevertheless, the connections are clear.

The above formulation considers a specific information pattern, in which the local sensors have access to all of the past summary messages of all sensors. Other information patterns have been considered elsewhere. These include the case in which the summary message u_{ni} depends only on Y_{ni} [50]; only on $Y_{ni}, u_{n(i-1)}, \dots, u_{n1}$ [52]; only on $Y_{ni}, Y_{n(i-1)}, \dots, Y_{n1}$ [49], [51]; or only on $Y_{ni}, Y_{n(i-1)}, \dots, Y_{n1}, I_{i-1}$ [52]. A discussion of optimality in these various situations is found in [52].

C. Distributed Detection with Sensors Performing Sequential Tests

In the preceding situation, the fusion center controlled the number of observations used by the distributed sensing network to make a decision. A second situation of interest is that in which the sensors themselves perform sequential tests; that is, each sensor decides, based on its own information pattern, when to stop taking observations. In this situation, there is no fusion center to couple the actions of the distributed sensors. However, the sensors act as a team to optimize a joint performance index. Thus, the sensors' actions are coupled through this common performance goal.

This problem has been considered in [53], [54], and [60]. The most general results are found in [54], and the basic idea can be summarized as follows. Suppose, as before, that there are N sensors, operating with independent streams of i.i.d. observations whose marginal distributions are governed by the validity of one of two possible statistical hypotheses. Each of the sensors will impose a stopping rule and a terminal decision rule, with the objective of minimizing (as a team) the performance index

$$E\{C(\delta_{\nu_1}, \delta_{\nu_2}, \dots, \delta_{\nu_N})\} + cE\{\max\{\nu_1, \nu_2, \dots, \nu_N\}\} \quad (35)$$

where

- ν_n is the stopping time of the n th sensor;
- δ_{ν_n} is the terminal decision rule of the n th sensor;

- $C(\cdot)$ is a function measuring the costs associated with various sets of decisions; and
- $c > 0$ is the cost per unit of observation delay.

Note that $C(\cdot)$ could correspond to the imposition of some fusion rule. For example, it might penalize errors only in the majority decision of the sensors. So, $C(\cdot)$ can be thought of as playing the role of a fusion center in coupling the sensors. Within some relatively mild conditions on the function $C(\cdot)$, it can be shown that the optimum sensor strategies are to perform tests similar to (24), but with time-varying boundaries. That is, the stopping time for the n th sensor is

$$\nu_n = \min \{i \geq 1 \mid \pi_i^{(n)} \notin (\pi_L^{(n)}(i), \pi_U^{(n)}(i))\} \quad (36)$$

where

$$\pi_i^{(n)} = P(H_1 \mid Y_{n1}, \dots, Y_{ni}), \quad i = 1, 2, \dots \quad (37)$$

and where $\pi_L^{(n)}(i)$ and $\pi_U^{(n)}(i)$ are deterministic sequences of thresholds. The lower thresholds $\pi_L^{(n)}(i)$ increase with i , and the upper thresholds $\pi_U^{(n)}(i)$ decrease with i . The computation of these thresholds is discussed in [54]. For each sensor n , the quantity (37) is computed as in (26) with L_i replaced by the likelihood ratio based on the i th sample of the n th sensor.

This problem can also be solved if the maximum stopping time in (35) is replaced with the sum of the stopping times (or a linear combination of the stopping times). In this case the optimum sensor decision rules will be of the same form as (36) except that the thresholds will not depend on the observation number i . (They will still depend on the sensor number n though.) This case (which was treated originally in [53]) corresponds to the situation in which there is a cost per observation but not a direct cost for decision delay. If both the number of observations and the maximum observation time are penalized, then the optimal strategies return to those with time-varying boundaries [54].

D. Distributed Quickest Detection

Another sequential-type detection problem of interest is that of *quickest detection*. Like ordinary sequential detection, quickest detection also involves the choice of a stopping rule. However, the basic hypotheses of interest in quickest detection are not binary, but rather there is one hypothesis for each possible observation time. In particular, the t th hypothesis is that the distribution of the observations changes at time t , for $t = 1, 2, \dots$.

To describe this problem, let us consider first the centralized situation. In this case we have a single observation stream $Y_1, Y_2, \dots$. Suppose there is an unknown time t such that, given t , the observations $Y_1, \dots, Y_{t-1}$ are i.i.d. with marginal distribution P_0 , and $Y_t, Y_{t+1}, \dots$ are i.i.d. with another marginal distribution P_1 . The quickest detection problem (also known as the *disorder* problem) is to detect such a change in distribution as soon as possible after it occurs, within some constraint on the rate or probability of false alarms.

There are two basic formulations of this problem, one in which the change point t is assumed to be a random variable with a known prior distribution, and the other in which t is assumed to be unknown but nonrandom (and it is possible that $t = \infty$). We will focus on the first of these formulations. In this case, we are interested in designing a stopping rule whose corresponding stopping time T minimizes the performance measure

$$P(T < t) + cE\{(T - t)^+\} \quad (38)$$

where $x^+ = \max\{x, 0\}$, and where $c > 0$ is a per-observation cost of delay. Note that $P(T < t)$ is the probability of announcing a change before it occurs—i.e., this is the probability of false alarm; whereas $E\{(T - t)^+\}$ is the average delay until the change is detected.

A particularly useful form of the quickest detection problem is the Shirayev–Roberts problem, in which the prior for t is a geometric distribution

$$P(t < 1) = q \quad \text{and} \quad P(t = k \mid t \geq k) = \rho, \quad k = 1, 2, \dots \quad (39)$$

where q and ρ are two parameters lying in the interval $(0, 1)$. The interpretation of this prior is that the conditional probability that the change occurs now, given that it has not occurred before now, is the constant ρ . The other constant, q , is the probability that the change has already occurred when we begin making observations. This is a common model for the occurrence of events in applications such as reliability and fault detection.

For the geometric prior (39), the stopping time minimizing the quantity (38) is [55]

$$\nu' = \min \{i \geq 1 \mid \pi'_i \geq \pi'_U\} \quad (40)$$

where $\pi'_U \in (0, 1)$ is an appropriately chosen threshold, and where

$$\pi'_i = P(t \leq i \mid Y_1, \dots, Y_i), \quad i = 1, 2, \dots \quad (41)$$

The sequence π'_i can be computed via the recursion

$$\pi'_i = \frac{[\pi'_{i-1} + (1 - \pi'_{i-1})\rho]L_i}{[\pi'_{i-1} + (1 - \pi'_{i-1})\rho]L_i + (1 - \pi'_{i-1})(1 - \rho)} \quad i = 1, 2, \dots \quad (42)$$

with $\pi_0 = q$ and where $L_i = L(Y_i)$ is the likelihood ratio between the prechange and postchange models based on the i th observation.

The distributed quickest detection problem has been considered by several authors, including [56]–[58]. For example, in [58], the Shirayev–Roberts formulation of the quickest detection problem is considered in the distributed sequential framework described in Section V-B above, in which the fusion center performs a quickest detection procedure on summary messages from the sensors. In this case, the optimal stopping time bears the same relationship to (40) that (30) does to (24). That is, the optimum stopping time is of the form

$$\tilde{\nu}' = \min \{i \geq 1 \mid \tilde{\pi}'_i \geq \tilde{\pi}'_U\} \quad (43)$$

where $\tilde{\pi}'$ is an appropriate threshold and where

$$\tilde{\pi}'_i = P(t \leq i | I_i), \quad i = 1, 2, \dots \quad (44)$$

[Here, I_i is defined as in (28).] The update rule for $\tilde{\pi}'$ is combination of those for $\tilde{\pi}$ and π' [i.e., (32) and (42)]. In particular

$$\tilde{\pi}'_i = \frac{[\tilde{\pi}'_{i-1} + (1 - \tilde{\pi}'_{i-1})\rho]\tilde{L}_i}{[\tilde{\pi}'_{i-1} + (1 - \tilde{\pi}'_{i-1})\rho]\tilde{L}_i + (1 - \tilde{\pi}'_{i-1})(1 - \rho)} \quad i = 1, 2, \dots \quad (45)$$

with $\tilde{L}_i$ defined as in (33) in terms of the local rules for calculating summary messages. As with the distributed sequential case, these latter rules are per-sample likelihood-ratio comparisons with thresholds depending on $\tilde{\pi}'_{i-1}$.

In addition to the formulation of Section V-B, a sensors-only distributed quickest detection framework similar to the sequential detection framework of Section V-C is considered in [57]. A further approach to this problem, in which the change point is assumed to be nonrandom, has been considered recently [56].

VI. CONCLUSION

We have discussed and reviewed results, mostly from relatively recent work, on distributed signal detection based on ideas that go beyond the fundamental starting points of distributed signal detection. In particular, we have examined extensions of distributed signal detection results established for simple-hypothesis testing under specific conditions such as of conditional independence. Our discussion has included locally optimum distributed detection of weak signals in non-Gaussian noise under conditions of sensor-to-sensor correlated signals, nonparametric, CFAR, and robust distributed detection, and sequential distributed detection.

There are a number of important topics that have not been discussed here due to length constraints. Some of these topics are discussed in Part I of this paper [1] and in [23]. For example, we have focused on binary hypothesis testing problems and on cases with binary sensor decisions. The reader is referred to [23] for a discussion of multi-bit sensor decision and M -ary hypothesis testing problems. We have assumed the best case scenario in many respects, fault-free processors with negligible communication delay and without errors in their communications. Cases with processor faults and channel errors are considered in [61]–[64]. In much of our analysis of fixed-sample-size schemes we have assumed that each sensor processor will generate decisions at periodic time instants. Some schemes without this restriction are discussed in [65] and [66]. We have considered very limited variations in our detection schemes over time. For example, adaptive thresholds were considered to provide constant false alarm probability in Section III. Much more general adaptive decision fusion schemes, some based on the theory of neural networks, have received attention lately [67]–[71]. We acknowledge that the reference list provided here is by no means exhaustive. There are other works which did not fit into the discussion that are on interesting specialized topics related to distributed signal detection.

Many interesting problems and scenarios remain to be investigated in distributed detection. These offer interesting mathematical and analytical challenges in addition to being of considerable interest in applications. Extensions of some of the work we have discussed to M -ary hypothesis testing problems should be pursued. Multiple hypothesis testing problems are of great interest, an example being the detection of M -ary symbols in communication systems. Analysis for detection system performance, under nonasymptotic conditions (finite number of sensors and observations, nonzero SNR) is also an important but challenging problem. The assumptions that have been made in obtaining the results so far, while useful and interesting, need to be relaxed further. For example, nonhomogeneous statistical distributions of observations across the sensors need to be taken into account in the type of schemes we have reviewed here (note that such nonhomogeneity is allowed in the sequential results described in Section V). Further work is also indicated in the synthesis of some of the techniques we have discussed above. Nonparametric and robust sequential distributed detection, for example, has received very little attention so far. Techniques such as conditional statistical tests and multivariate procedures (for nonparametric hypotheses) could also be applied very fruitfully in distributed signal detection applications.

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Rick S. Blum (Member, IEEE) received the B.S. degree in electrical engineering from Pennsylvania State University, University Park, and the M.S. and Ph.D. degrees in electrical engineering from the University of Pennsylvania, Philadelphia, in 1984, 1987, and 1981, respectively.

From 1984 to 1991 he was a member of the Technical Staff at General Electric Aerospace in Valley Forge, PA. Since 1991 he has been an Assistant Professor in the Electrical Engineering and Computer Science Department at Lehigh University in Bethlehem, PA. His research interests include signal detection and estimation and related topics in the areas of signal processing and communications. He holds one patent for a parallel signal and image processor architecture.

Dr. Blum is a member of Eta Kappa Nu and Sigma Xi.



Saleem A. Kassam (Fellow, IEEE) received the B.S. degree in engineering from Swarthmore College, Swarthmore, PA, the M.S.E., M.A., and Ph.D. degrees in electrical engineering from Princeton University, Princeton, NJ, in 1972, 1974, and 1975, respectively.

Since 1975 he has been a Professor at the Moore School of Electrical Engineering, University of Pennsylvania, Philadelphia. He served as Chairman of the Department of Electrical Engineering from 1992 to 1994. His research interests are in the area of statistical signal and array processing and communications, and he has made contributions to signal detection theory, robust signal processing, nonlinear and adaptive filtering, image processing, spectrum estimation, spatial array processing, and microwave and acoustic imaging. He co-edited a book on nonparametric detection and is the author of *Signal Detection in Non-Gaussian Noise* (Springer-Verlag, 1988). He holds three patents on array imaging systems. He was an Associate Editor for the IEEE TRANSACTIONS ON INFORMATION THEORY during 1990–1992.

Dr. Kassam is a member of Phi Beta Kappa and Sigma Xi.



H. Vincent Poor (Fellow, IEEE) received the Ph.D. degree from Princeton University, Princeton, NJ, in 1977.

Since 1990 he has been a Professor of Electrical Engineering at Princeton University. From 1977 to 1990 he was a faculty member at the University of Illinois. He has also held visiting positions at several universities and research organizations in the United States, United Kingdom, and Australia. His research interests are in statistical signal processing and its applications, and his publications in this area include the graduate textbook *Introduction of Signal Detection and Estimation* (Springer-Verlag, 1988 and 1994).

Dr. Poor is a Fellow of the Acoustical Society of America and the American Association for the Advancement of Science. He served as President of the IEEE Information Theory Society in 1990 and a member of the Board of Directors in 1991 and 1992. He received the 1991 Terman Award from the American Society for Engineering Education and the 1994 Distinguished Member Award from the IEEE Control Systems Society.